

**Vijayalakshmi Jakkati**  
**Machine Learning / Artificial Intelligence Engineer**  
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## PROFILE SEEK

Artificial Intelligence Engineer | Machine Learning Engineer | MLOP Engineer |

## SUMMARY

- **AI/ML Engineer with 5 years of experience** building scalable machine learning systems, **GenAI applications**, data pipelines, and full-stack AI- powered applications across manufacturing, geospatial, and marketing domains.
- Proficient in **Python, SQL, and ML frameworks** including scikit-learn, XGBoost, TensorFlow, and PyTorch, with strong skills in data wrangling, model evaluation, and deployment.
- Experienced in developing and deploying **LLM-powered features** using **OpenAI GPT, Salesforce BLIP, and transformers** for real-time prediction, explanation, and content generation.
- Built **BizDash.ai**, a full-stack dashboard using React, Django, and AWS (EC2, RDS, S3, ECS Fargate), enabling real-time data ingestion and visualization, reducing reporting cycles by 55%.
- Delivered Location Finder, a **GenAI-enhanced geolocation platform** using **MobileNetV2, BLIP, and GPT-3.5**, achieving 92% image prediction accuracy and automating distance/cost estimation.
- Delivered a **recommendation system** for Cabot Corporation using content/collaborative filtering and **Bayesian Optimization**, improving formulation accuracy by 20% and reducing R&D time by 30%.
- Proficient in **natural language processing (NLP)** techniques including **chatbot development**, sentiment analysis, and text generation using transformer models like GPT, BERT, and BLOOM.
- Automated enterprise ETL workflows with **Oracle Data Integrator (ODI)** and integrated cross-functional tools (ERP, Salesforce), improving data flow reliability and system uptime.
- Hands-on experience with **Docker, GitHub Actions, and cloud deployments (AWS, Azure)**, backed by agile project management with **Jira**, CI/CD pipelines, and API testing with **Postman**.

## TECHNICAL SKILLS

- **Programming Language:** Python (Pandas, NumPy, scikit-learn), R, SQL, C++, Core JAVA
- **Databases:** Microsoft SQL Server, MySQL, Oracle 12c
- **Data Analysis and Visualization:** Tableau, Matplotlib, Power BI
- **Machine Learning:** Regression, Classification, Clustering, Neural Networks, CNNs, Transformers, Model Evaluation, Hyperparameter Tuning
- **Data Science:** GenAI,NLP,Data Cleaning,Preprocessing, and Wrangling,Exploratory Data Analysis (EDA), Feature Engineering and Selection.
- **Cloud Technologies:** Azure, Aws
- **Deployment and product ionization:** Docker, Kubernetes, Django, Rest API, React
- **Development Tools:** Git and GitHub
- **ETL Tools & Methodologies:** ODI, OBIEE, DBMS
- **Deep Learning Frameworks:** PyTorch, TensorFlow, Keras
- **Reporting Tools:** MS Office (Word/Excel/PowerPoint/ Visio/Outlook), OBIEE Reports, Service Now, Jira

## EDUCATION

**University of Central Missouri (GPA: 3.8/4.0), Missouri, USA**

*Master of Science in Computers; Concentration in Computer Science 2024*

**NRI Institute of Technology, Nuzvid, India (CGPA: 8.01/10), India**

*Bachelor of Engineering; Concentration in Electronics & communications, India 2020*

## PROFESSIONAL CERTIFICATIONS

**OCI (Oracle Cloud Infrastructure) & ODI 2021**

**Data Analytics and Visualization by Accenture (Forage) 2023**

**Power BI by PWC by Forage 2023**

## WORK EXPERIENCE

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**RealCoderz**  
**Machine Learning / Artificial Intelligence Engineer**

**Reston, VA USA**  
**Aug 2024 – Present**

### **Location Finder – Full-Stack AI-Powered Location Intelligence Platform**

Location Finder is a full-stack AI-powered web platform that enables users to upload an image of a global landmark and receive intelligent predictions including the landmark name, coordinates, distance from home, estimated travel cost, and rich natural language descriptions generated by **GenAI (Generative AI) and LLMs**. The system combines deep learning models (MobileNetV2, BLIP), OpenAI GPT integration, and a robust Django & React architecture to deliver real-time geolocation intelligence with GenAI (Generative AI) explainability.

**Problem Solved:** Manual identification and tagging of landmarks from photos are time-consuming, error-prone, and lacks contextual richness. The project automated this process with AI-driven image classification and captioning, improving detection accuracy by 92%, reducing lookup time by 80%, and enabling contextual explanations via GPT. The platform introduced geospatial awareness, eliminating manual distance calculations and improving user engagement with multi-role access control.

#### **Key Responsibilities:**

- **Architecture & Design:** Designed a scalable full-stack system using React (Vite) and **Django REST Framework**, implementing secure **JWT-based authentication** and role-based access control to streamline user management.
- **Frontend Development with React:** Built responsive UI with protected routes, dark mode, and real-time updates using **Context API and Toastify**, improving UX efficiency by 40%.
- **Backend Development with Django:** Developed **RESTful APIs** for image upload, location management, and user permissions. Integrated JWT (SimpleJWT) for secure, token-based authentication with full test coverage.
- **AI-Powered Landmark Prediction & Geospatial Intelligence:** Trained a MobileNetV2 model with transfer learning for landmark classification (92% accuracy) and integrated GeoPy + Nominatim API for distance and travel cost estimation.
- **Image Captioning using GenAI and LLMs (BLIP + GPT-3.5 Turbo):** Leveraged **Salesforce BLIP** for visual recognition and integrated **OpenAI's GPT-3.5 Turbo (LLM)** to generate rich, multi-line captions. This **LLM-driven GenAI pipeline** enhanced engagement by 60% and improved contextual understanding of image predictions.
- **NLP-Powered Captioning and Explanation:** Integrated natural language processing (**NLP**) capabilities using BLIP and GPT-3.5 Turbo to generate rich, context-aware captions for landmarks. This NLP pipeline enabled dynamic natural language generation and improved user interpretation of visual predictions.
- **Version Control & CI/CD:** Managed multi-repo structure with **GitHub** and automated **testing/build pipelines** using **GitHub Actions** for both frontend and backend deployments.
- **Azure Cloud Deployment & Automation:** Deployed backend via Azure App Service, stored media in Blob Storage, containerized with Docker and orchestrated using AKS; monitored via Azure Monitor & Log Analytics.

**Technologies Used:** React.js, Django, MySQL, Docker, Azure (App Service, Blob Storage, ACR, AKS, Monitor), GitHub Actions, MobileNetV2, GPT-3.5 Turbo, GenAI, LLM, Salesforce BLIP, GeoPy, Postman, Jira

**Cognizant Technologies Solutions,**  
**Data Scientist**

**Hyderabad, India**  
**Feb 2021 - Jul 2023**

### **Intelligent Product Formulation Recommendation System – Specialty Chemicals Domain**

#### **Problem Solved:**

Cabot's global operations in the specialty chemicals industry required optimization of product formulations across domains like automotive, electronics, and energy storage. Traditional R&D processes were manual and time-consuming, while backend operations faced fragmented ETL workflows and inconsistent data synchronization across ERP and CRM systems. This project aimed to streamline formulation development using machine learning and improve enterprise data reliability through automation and integration.

#### **Key Responsibilities & Achievements:**

- **Data Aggregation & Preprocessing:** Consolidated historical formulation data, performance metrics, and customer feedback using **Python (Pandas, NumPy)** and **SQL**. Standardized and cleaned datasets by addressing missing values

and outliers, creating a robust foundation for analysis.

- **Feature Engineering & Hybrid Recommendation Engine:** Built features to model chemical component interactions and developed a hybrid recommendation system using content-based filtering (chemical attributes) and collaborative filtering (client feedback). Trained models using scikit-learn and XGBoost, achieving a 20% improvement in formulation success rate.
- **Optimization & Deployment:** Used Bayesian Optimization to identify optimal component ratios, significantly reducing trial-and-error iterations. Deployed the solution as a RESTful API using FastAPI, enabling integration with internal R&D platforms.
- **Impact & Business Value:** Reduced formulation development time by **30%**, accelerated product delivery, and improved client satisfaction—resulting in a **15% increase in repeat business**.
- **NLP for R&D Documentation Insights:** Applied **natural language processing (NLP)** to analyze unstructured R&D logs and historical formulation notes. Extracted relevant patterns and chemical interactions to enrich the feature set, improving model accuracy and interpretability.
- **Enterprise Data Integration & ETL Automation:** Automated ETL pipelines using **Oracle Data Integrator (ODI SDK scripting)**, reducing manual effort by over **15 hours/month** and improving cross-system data consistency by **25%**. Integrated third-party tools like **Oracle ERP** and **Salesforce** to unify business reporting and analytics.
- **System Monitoring & Incident Management:** Maintained **99.9% system uptime** through proactive monitoring and troubleshooting. Delivered Tier 2 support for **500+ users**, achieving **90% first-call resolution** and managing over **1,000 annual support tickets** with a **95% satisfaction rate**.

**Technologies Used:** Python, Pandas, NumPy, SQL, scikit-learn, XGBoost, FastAPI, Oracle Data Integrator (ODI), Oracle ERP, Salesforce, REST APIs, Bayesian Optimization, ServiceNow

**Veracity Software Inc,**  
**Data Science Internship**

**Pune, India**  
**May 2019 – Dec 2020**

### **Data Analysis & Predictive Modeling – Sales & Marketing Analytics**

Developed end-to-end predictive analytics solutions to support targeted marketing and sales forecasting strategies. The project unified fragmented data pipelines and improved visibility into customer behavior, significantly boosting campaign performance and reporting efficiency.

- **Problem Solved:** Marketing and sales teams lacked visibility into customer behavior due to siloed data sources and manual reporting processes. This led to ineffective targeting, slow decision-making, and underperforming campaigns. The solution centralized data pipelines, introduced real-time analytics, and enabled predictive modeling to improve marketing outcomes and business forecasting accuracy.
- **Project Development:** Designed and implemented data workflows from collection and preprocessing to model development and visualization. Collaborated closely with business units to align analytics outputs with operational KPIs and campaign goals.
- **Data Integration:** Unified datasets from multiple systems (sales, CRM, campaign data) using SQL and Azure Data Factory, ensuring consistency across sources and enabling comprehensive analysis.
- **Model Development:** Performed EDA to uncover trends, churn patterns, and performance gaps. Built regression, classification, and clustering models using scikit-learn, leading to a 20% increase in campaign ROI.
- **Dashboard Design:** Created interactive dashboards in Power BI and Tableau to visualize KPIs, customer segments, and regional sales trends. Improved stakeholder engagement by 25% through intuitive, real-time reporting.
- **Technical Optimization:** Optimized SQL queries and views, reducing report generation times by 50%. Applied feature engineering techniques to enhance model performance and reliability.
- **Insights Delivery:** Translated model outputs and dashboard insights into actionable strategies for marketing, finance, and operations teams—enhancing collaboration and data literacy across departments.

**Technologies Used:** Python (Pandas, NumPy), scikit-learn, SQL, Azure Data Factory, Power BI, Tableau, Jupyter Notebook

### **ACCOMPLISHMENTS**

- Received Excellence in Delivery award for delivering complex work orders without any defects at Cognizant.

### **BizDash.ai – Full-Stack Enterprise Dashboard:**

- Built and deployed a full-stack dashboard application for automating business data ingestion, editing, and visualization.
- Designed a responsive frontend **using React.js, MUI, and Recharts** for dynamic file upload, editable tables, and real-time charts.
- Developed RESTful APIs with **Django** to support Excel/SQL ingestion, schema-based form generation, and chart data feeds.
- Automated background data generation using **Python scripts** to simulate real-time workloads and improve test coverage.
- Deployed backend on **AWS EC2**, managed media files with **S3**, used **RDS (MySQL)**, and containerized services with **Docker** and **ECS Fargate**.
- Implemented CI/CD with **GitHub Actions**, monitored system performance using **CloudWatch**, and followed **Agile practices with Jira**

### **Web-Based Kanban Board Application:**

- Developed and deployed a dynamic Kanban board application to enhance task management for **Agile and DevOps teams**.
- Enabled customizable boards, drag-and-drop functionality, secure authentication, and seamless collaboration.
- Built a responsive front-end **with React.js** and integrated it with a **Spring Boot backend**.
- Deployed the application using **AWS services like S3 for static hosting, EC2 for the backend, RDS for MySQL** database, and IAM for secure access.
- Implemented **CI/CD pipelines** with **GitHub Actions** for automated deployment.
- Conducted performance optimization, continuous monitoring, and logging to resolve issues effectively.

### **Sales Insights Dashboard Using Tableau:**

- Developed interactive dashboards to visualize sales trends, regional performance, and product profitability, boosting revenue by 10%.
- Automated data refresh process, reducing manual reporting efforts by 30%.
- Leveraged advanced **Tableau features like calculated fields**, Parameters and customs filters to deliver actionable insights.
- Integrated and processed datasets from **SQL Server** and Excel for real-time **dynamic Visualizations**.

## **GRADUATE COURSES**

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Advanced Data Base Systems | Algorithms | Compiler & Design | Data Visualization using R | Intro to Cloud Computing | Machine Learning | Object Oriented Programming System & Data structures | Application Programming in C# | Operating Systems | Prog Found Data Science & AI